MIL-C-20327
21 November 1951

MILITARY SPECIFICATION

CHARGES, PROPELLING, FOR SEPARATE LOADING AND SEMI-FIXED AMMUNITION

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-63-1C
3 May 1945

This specification consists of this cover sheet and Specification 50-63-1C, dated 3 May 1945, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-63-1C
3 May 1945
Superseding
No. 50-63-1B
20 June 1940

CHARGES, PROPELLING, FOR SEPARATE LOADING AND SEMI-FIXED AMMUNITION

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department drawing pertaining to the propelling charge under contract and all drawings and specifications listed or referred to thereon form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPES.

B-1. This specification covers all types of propelling charges for separate loading and semi-fixed ammunition.

C. MATERIAL AND WORKMANSHIP.

C-1. Materials and parts shall be in accordance with applicable specifications and drawings.

C-2. All work in connection with the assembly of separate loading and semi-fixed ammunition shall be performed in suitable buildings kept at a uniform temperature, which shall not be sufficiently below the prevailing outside temperature that condensation of moisture will occur.

C-3. When dyeing of cartridge cloth or igniter cloth is required, this shall be accomplished in a neutral bath. The bath will be considered neutral if the salt solution before the dye is added fails to change the color of neutral litmus paper within 30 seconds.

D. GENERAL REQUIREMENTS.

D-1. The moisture content of the black powder in the loaded igniter pad shall not exceed 0.70%.

E. DETAIL REQUIREMENTS.

E-1. The propelling charge shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Igniter charge for separate loading propelling charge.- The weight of the igniter charge shall not vary from that prescribed on the applicable drawings by more than plus or minus 0.10 ounce for charges weighing less than 8 ounces, nor more than plus or minus 0.20 ounce for charges of 8 ounces and over. The charges shall be weighed, or volumetrically measured, and at least 10 percent of the charges shall be checkweighed. (See Par F-1d.)

E-3. Propelling charge.- The propelling charge shall be weighed and checkweighed and the weight shall be as prescribed in the loading authorization within the tolerances prescribed in Table I. (See Par F-1e)

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TABLE I

Weapon	Unit of charge weight	Nominal[1] weight of charge unit	Tolerance (plus or minus)
		Pounds	Ounces
6 in. Gun, 1900, 1903, 1905	Base section	15.44	0.30
6 in. Gun, 1900, 1903, 1905	Increment section	11.00	.30
6 in. Gun, 1900, 1903, 1905	Single section	32.00	.50
6 in. Gun, 1897M1, 1908	Single section	29.00	.50
155 mm. Gun, 1917, 1917A1, 1918M1	Base section	21.00	.40
155 mm. Gun, 1917, 1917A1, 1918M1	Increment section	5.20	.10
155 mm. Gun, M1	Base section	22.00	.40
155 mm. Gun, M1	Increment section	10.25	.30
155 mm. How., 1917, 1917A1, 1918	Base section, zone 1	1.78	.02
155 mm. How., 1917, 1917A1, 1918	Increment 2, M1A1 charge	.27	.01
155 mm. How., 1917, 1917A1, 1918	Increment 3, M1A1 charge	.47	.01
155 mm. How., 1917, 1917A1, 1918	Increment 4, M1A1 charge	.58	.01
155 mm. How., 1917, 1917A1, 1918	Increment 5, M1A1 charge	.90	.01
8 in. Gun, 1888, 1888M1, 1888MII	Single section	82.38	1.60
8 in. Gun, 1888, 1888M1, 1888MII	2 section	42.25	1.60
8 in. Gun, 1888, 1888M1, 1888MII	Base section	50.00	1.60
8 in. Gun, 1888, 1888M1, 1888MII	Increment section	20.63	.40
8 in. Gun, Mk. VI, Mod. 3A2	Base section	70.00	1.60
8 in. Gun, Mk. VI, Mod. 3A2	Increment section	35.00	.50
8 in. How., 1917 Mk. VI	Base section	4.50	.10
8 in. How., 1917 Mk. VI	Increment 2	1.03	.05
8 in. How., 1917 Mk. VI	Increment 3	1.50	.05
8 in. How., 1917 Mk. VI	Increment 4	3.72	.10
8 in. How., Mk VII, VIII½	Increment, aliquot part	4.18	.10
240 mm. How., 1918	Increment, aliquot part	3.66	.10
10 in. Gun, 1888, 1888M1, 1888MII, 1895, 1895M1.	Base section	120.00	2.70
10 in. Gun, 1888, 1888MI, 1888MII, 1895, 1895MI.	Increment section	40.00	2.10
10 in. Gun, 1888, 1888MI, 1888MII, 1895, 1895MI.	Section	38.75	2.10
10 in. Gun 1900	Section	58.00	2.70
12 in. Gun, 1888, 1888M1, 1888MI½, 1888MII, 1895, 1895M1.	Section	67.50	2.30
12 in. Gun, 1895	Base section	150.00	2.70
12 in. Gun, 1895	Increment section	70.00	2.30
12 in. Gun, M1900	Section	73.75	2.30
12 in. Mort. 1890, 1890M1, 1908	Increment, aliquot part	6.41	0.20
12 in. Mort. 1912	Zone 4 section, M5 Chg	33.00	0.60
12 in. Mort. 1912	Increment 5, M5 Charge	5.25	0.20
12 in. Mort. 1912	Increment 6, M5 Charge	6.75	0.20
12 in. Mort. M1912	Increment 7, M5 Charge	9.81	0.30
12 in. Mort. M1912	Increment 8a, M5 Charge	11.94	0.40
	Zone 8b Section, M6 Chg.	59.87	0.70
	Increment 9, M6 Charge	12.81	0.40
	Increment 10, M6 Chg.	18.56	0.50

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Weapon	Unit of charge weighed	Nominal[1] weight of charge: Unit	Tolerance (plus or minus)
		Pounds	Ounces
14 in. Gun, 1907, 1907M1	Section	83.00	3.30
14 in. Gun, 1920M1 and MII	Section	115.00	3.30
14 in. Gun, M1920M1 and MII (1,200 pound proj.)	Section	125.00	3.30
14 in. Gun, 1910, 1910M1	Section	108.25	3.30
14 in. Gun, 1909	Section	217.50	5.40
16 in. Gun, 1919, 1919MII, 1919MIII	Section	104.00	3.30
16 in. Gun, 1919, 1919MII, 1919MIII	Section	208.00	5.40
16 in. Gun, 1895	Single section	650.00	6.00
16 in. Gun, Mk. II (Navy)	Section	112.00	3.30
16 in. How. 1920	Base section	176.00	5.40
16 in. How. 1920	Increment section	40.00	2.10
75 mm. Pack Howitzer, M1	Base section Zone 1	0.37	0.01
75 mm. Pack Howitzer, M1	Increment 2	0.11	0.01
75 mm. Pack Howitzer, M1	Increment 3	0.14	0.01
75 mm. Pack Howitzer, M1	Increment 4	0.30	0.01
105 mm. Howitzer, M2	Base Section, Zone 1	0.68	0.01
105 mm. Howitzer, M2	Increment 2	0.13	0.01
105 mm. Howitzer, M2	Increment 3	0.16	0.01
105 mm. Howitzer, M2	Increment 4	0.21	0.01
105 mm. Howitzer, M2	Increment 5	0.34	0.01
105 mm. Howitzer, M2	Increment 6	0.54	0.01
105 mm. Howitzer, M2	Increment 7	0.86	0.01
105 mm. Howitzer, M2	Charge for M67 Shell	1.50	0.02
105 mm. Howitzer, M3	Base Section, Zone 1	0.62	0.01
105 mm. Howitzer, M3	Increment 2	0.10	0.01
105 mm. Howitzer, M3	Increment 3	0.13	0.01
105 mm. Howitzer, M3	Increment 4	0.19	0.01
105 mm. Howitzer, M3	Increment 5	0.31	0.01
105 mm. Howitzer, M3	Charge for M67 Shell	1.20	0.02
4.5 in. Gun, M1	Base Section	7.76	0.20
4.5 in. Gun, M1	Increment	3.32	0.10
155 mm. Howitzer, M1	Zone 1 Sec. M3 Chg.	1.96	0.02
155 mm. Howitzer, M1	Increment 2, M3 Chg.	0.49	0.01
155 mm. Howitzer, M1	Increment 3, M3 Chg.	0.66	0.01
155 mm. Howitzer, M1	Increment 4, M3 Chg.	0.92	0.01
155 mm. Howitzer, M1	Increment 5, M3 Chg.	1.61	0.02
155 mm. Howitzer, 1917 1917A1, 1918	Zone 3, Sec M2A1 Chg.	3.94	0.05
155 mm. Howitzer, 1917 1917A1, 1918	Increment 4, M2A1 Chg.	0.75	0.01
155 mm. Howitzer, 1917 1917A1, 1918	Increment 5, M2A1 Chg.	0.88	0.01
155 mm. Howitzer, 1917 1917A1, 1918	Increment 6, M2A1 Chg.	1.94	0.02
155 mm. Howitzer, 1917 1917A1, 1918	Increment 7, M2A1 Chg.	0.81	0.01
155 mm. Howitzer, M1	Zone 3, Section, M4A1 Chg.	3.96	0.05
155 mm. Howitzer, M1	Increment 4, M4A1 Chg.	1.16	0.02
155 mm. Howitzer, M1	Increment 5, M4A1 Chg.	1.73	0.02
155 mm. Howitzer, M1	Increment 6, M4A1 Chg.	2.79	0.08
155 mm. Howitzer, M1	Increment 7, M4A1 Chg.	3.42	0.05

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Weapon	Unit of charge weighed	Nominal[1] weight of charge: Unit	Tolerance (plus or minus)
		Pounds	Ounces
8 in. Gun, M1	Base section, M9 Charge	53.00	0.50
8 in. Gun, M1	Increment, M9 Charge	21.50	0.30
8 in. Gun, M1	Base section, M10 Charge	86.69	1.60
8 in. Gun, M1	Increment, M10 Charge	11.44	0.50
8 in. Howitzer, M1	Zone 1 Section, M1 Charge	4.41	0.10
8 in. Howitzer, M1	Increment 2, M1 Charge	1.36	0.05
8 in. Howitzer, M1	Increment 3, M1 Charge	1.91	0.05
8 in. Howitzer, M1	Increment 4, M1 Charge	2.56	0.05
8 in. Howitzer, M1	Increment 5, M1 Charge	3.11	0.10
8 in. Howitzer, M1	Zone 5 Section, M2 Charge	16.50	0.30
8 in. Howitzer, M1	Increment 6, M2 Charge	5.22	0.10
8 in. Howitzer, M1	Increment 7, M2 Charge	6.22	0.10
240 mm. Howitzer, M1	Base Section, Zone 1	43.22	0.50
240 mm. Howitzer, M1	Increment 2	10.45	0.20
240 mm. Howitzer, M1	Increment 3	12.74	0.20
240 mm. Howitzer, M1	Increment 4	13.25	0.20

[1]The prescribed wieghts of charge vary with different lots of powder and are given on the loading authorization for the particular lot.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection.-

F-1a. Inspection shall be as specified in United States Army General Specification for Ammunition Except Small Arms Ammunition, No. 50-0-1.

F-1b. The inspector shall assure himself that the bags are filled with the proper kind, grade, and quality of propelling charge, and that the bags are made in accordance with the drawings and specifications.

F-1c. Gaging separate loading propelling charge.- After the propelling charge has been wrapped or tied as tightly as possible, the contractor shall gage each charge in a diameter and length gage pertaining to the charge under contract. If the charge fails to enter the diameter and length gage freely, or if the gage indicates that the charge is of improper length, it shall be rejected, subject to correction and retest. Drawings of the Government diameter and length gage will be furnished to the contractor for his information upon application to the inspector.

F-1e. Propelling charge.- The inspector shall assure himself that the contractor has weighed and checkweighed the propelling charge and that the requirement as to weight has been met. The checkweighing shall be accomplished by an operator working independently of the original weigher, using a different scale from that use to make the first weighing.

F-1f. Inspector's checkweighing.- If the inspector decides to weigh any of the propelling charges to assure himself that the requirements as to weight are met, his weighing shall, in general, be done by weighing assembled charges or sections and deducting the average weight of bag and igniter or he may break down assembled charges or sections of charges from each lot of powder and weigh the powder separately

F-2. Lots.- There are no ammunition lot numbers assigned to propelling charges for separate loading ammuntion.

G. PACKAGING, PACKING, AND MARKING.

G-1. Packing.-

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G-1a. As specified on the drawing for the particular charge being loaded.

G-1b. When metal cartridge storage cases are used, they shall be free from grease, rust, sand, chips, and other foreign material, and shall have the interior surface dry and uniformly coated, as required by the cartridge storage case drawing. If examination shows that the interior wall is uncoated, imperfectly coated, or contaminated in any way, the cases shall be cleaned and, where necessary, coated so that the entire inside surface complies with the cartridge storage case drawing. In no case shall any acid or alkali be used in the cleaning process for the interior of the storage cases.

G-1c. The painting and marking shall be restored by the contractor on any cartridge storage cases furnished to him painted and marked when such painting and markings have been marred or obliterated.

G-1d. After placing the charge in the cartridge storage case, all metal cases shall withstand, without leakage, an air pressure between 2 and 3 pounds per square inch for a period of not less than 1 minute. Fiber cases will not require an air pressure test.

G-1e. Data card for Separate Loading Propelling Charge.- Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of charges in accordance with the form specified below. Ten of these cards completely filled out for each lot shall accompany each shipping ticket, and seven cards completely filled out shall be furnished the inspector.

AMMUNITION DATA TAG FOR SEPARATE LOADING PROPELLING CHARGE

Name and place of manufacture.....
Date..... Order No.....
(Full or fraction) of charge..... For.....
(Size and model of gun in which used)
Kind of propellent..... Mfg.[1] Lot No..... of year.....
Net weight[2].....
Igniter: Ind of powder..... Granulation.....
Weight..... Distribution .....

BALLISTIC DATA

Weight of full charge, net.....pounds.....ounces
Weight of projectile .....pounds
Muzzle velocity ..... feet per second for model ..... gun
Pressure .... lbs. per sq. in.
(Specify any other guns, projectiles that this charge can be used with at bottom of tag.)

Certified to by .....Inspector
(Print here name of inspector, district, arsenal, or loading plant, and sub-district, where applicable.)

[1]Manufacturer's initials denoting factory, as "P.A." for Picatinny Arsenal, "I.O.W." for Indiana Ordnance Works, etc.
[2]Net weight is required only for sectional charges.

G-1f. Ammunition Data Card for Semi-Fixed Propelling Charge.- Data cards 5 x 8 inches, made of commercial tag board, approximately 200 pounds ream weight, shall be prepared for each lot of charges in accordance with the form specified below. Two of these cards completely filled out for each lot shall accompany each shipping ticket and six cards completely filled out shall be

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furnished the inspector.

AMMUNITION DATA CARD FOR SEMI-FIXED PROPELLING CHARGE

Kind:.....
Loaded and Assembled by:..... Date .....
Order No.....
Packed:.....
Remarks.....

BALLISTIC DATA

Zone	Wt. of Chg. by Zone Ounces	Total Wt. of Chg. Ozs.	Muzzle Velocity (Ft./Sec.)	Pressure (Lbs./sq.in.)

Certified to be .....Inspector
(Print here name of inspector, district, arsenal, or loading plant, and sub-district where applicable.)

G-2. Marking.- The marking for shipment shall be as specified on the drawings for the particular propelling charge.

H. NOTES

H-1. A propelling charge for separate loading ammunition is that assembly of propellent powder and igniter charge in cloth bags to form zones or sections for use in weapons in which the projectile, propelling charge, and primer are not assembled as a unit before loading in the weapon.

H-2. A propelling charge for semi-fixed ammunition is that assembly of propellent powder in cloth bags for use in a cartridge case as a component of the complete round. The projectile can be removed for the purpose of reducing propelling charge by removing various sections before loading in the weapon.

H-3. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

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